USN | | | | | | | | | | [22CSE532]

FIFTH SEMESTER B. TECH EXAMINATION, DECEMBER-2024

ADVANCED MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5x2=10**
1.	**a.**	What is meant by TensorFlow?	L1	CO1	**02**
	b.	Summarize the ensemble learning methods.	L1	CO2	**02**
	c.	Sketch the block diagram for Policy based methods in RL	L2	CO3	**02**
	d.	Compare model validation and testing.	L2	CO4	**02**
	e.	List out the types of web-development architecture.	L1	CO6	**02**
		PART - B			**6x5=30**
2.	**a.**	Explain the differences between supervised and unsupervised learning.	L2	CO1	**05**
		OR			
	b.	Explain multiple layers TensorFlow.	L2	CO1	**05**
3.	**a.**	Discuss in detail the features of TensorFlow.	L2	CO1	**05**
		OR			
	b.	Discuss in detail the disadvantages of TensorFlow.	L2	CO1	**05**
4.	**a.**	Explain boosting and Ada-boosting algorithm with a neat sketch.	L2	CO2	**05**
		OR			
	b.	Explain Random Forest with example.	L2	CO2	**05**
5.	**a.**	Discuss in brief Q learning.	L2	CO3	**05**
		OR			
	b.	Explain the key terms in reinforcement learning.	L2	CO3	**05**
6.	**a.**	Write short notes on Bootstrapping and Jackknife.	L2	CO4	**05**
		OR			
	b.	Discuss in detail about Hold-out validation.	L2	CO4	**05**
7.	**a.**	Discuss the different ways to deploy the Machine Learning model.	L2	CO6	**05**
		OR			
	b.	What is meant by AWS? Explain its components functionality and features.	L2	CO6	**05**
		PART - C			**6x10=60**
8.	**a.**	Discuss Backpropagation rule considering the training rule for output unit weights and training rule for hidden unit weights with example.	L2	CO1	**10**
		OR			

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	b.	Discuss the functioning of the forward propagation algorithm with examples.	L2	CO1	**10**
9.	a.	Illustrate ensemble learning methods algorithms with examples.	L3	CO2	**10**
		OR			
	b.	Illustrate bagging approach with example.	L3	CO2	**10**
10.	a.	Write short note on value-based methods in RL.	L2	CO3	**10**
		OR			
	b.	Discuss model-based methods in RL with examples.	L2	CO3	**10**
11.	a.	Discuss in detail Deep Q-Learning with suitable example.	L2	CO3	**10**
		OR			
	b.	Explain fitted Q-learning process with an example. Write the steps of the algorithm. Also discuss how it is different from Q-learning.	L2	CO3	**10**
12.	a.	Describe briefly pipeline and illustrate how pipelines can be helpful in data science to streamline the workflow of a project.	L2	CO4	**10**
		OR			
	b.	Illustrate Regression metrics and Ranking metrics with examples.	L2	CO4	**10**
13.	a.	Discuss web-development architecture and its types. Also differentiate between MVC and MVP architecture.	L2	CO6	**10**
		OR			
	b.	Compare Django frame work with flash frame work.	L2	CO6	**10**